Case study brief

Partner and campaign performance

From	Head of Marketing and Partnerships
To	Data and Analytics team
Subject	Which referral partners and campaigns are worth the money?
Scenario	Fictional South African consumer lender. All data is simulated (Nov 2024 to Oct 2026).
Due	Two weeks from kickoff

Background

We source personal loans through our own channels (app, branch, website) and through 9 paid referral partners. Each partner earns a commission on every loan we disburse, and we also fund their marketing campaigns. We have never checked whether that spend pays off. Partners are currently judged by how many applications they send.

The request

Tell us which partners and campaigns are worth the money, which are not, and what we should do about it. A partner that sends many loans that go bad is not a good partner, so we want the answer in terms of money and not volume.

What we need back

- A partner scorecard: applications, approval rate, loans, default rate, commission, campaign spend and margin per rand lent.
- A ranking of campaigns by cost per approved loan.
- A statement of how sure you are, including whether the ranking changes if your assumptions change.
- A one-page recommendation: where to spend more, where to cut, what to renegotiate.
- Notebooks we can re-run.

Data provided

The existing customer, application, loan, repayment and partner tables (cleaned and loaded by the data pipeline project), plus a new file that the pipeline also cleans:

File	One row per	Notes
campaigns.csv	marketing campaign	Partner, name, start and end date, spend in rand. Entered by hand in a spreadsheet, so the raw file is not clean; the pipeline cleans it and links each application to the campaign running for its partner.
partners.csv	referral partner	Channel type and commission percentage (of principal).

Applications record the partner but not the campaign. Direct channels have no campaigns.

Constraints

- State every assumption (for example how much of the principal is lost when a loan defaults) and show whether the answer depends on it.

- Do not claim a campaign caused extra applications. We can only measure what happened while it ran.
- Use only the platform we have: Databricks, Python, SQL and PySpark.

Out of scope

Pricing, changes to commission contracts (we will decide those from your recommendation), and testing campaigns that have not run yet.

Questions to settle at kickoff

Question	Working assumption used in this case study
Which campaign gets an application?	The campaign its partner was running on the application date. If two overlap, the one that started most recently.
What is margin?	Interest income, minus credit loss, minus commission, minus campaign spend, per R1,000 lent.
What does a default cost?	A defaulting loan pays no interest and loses 60% of the principal. Re-tested at 40% and 80%.
What about unknown spend?	Left out of cost calculations, not counted as zero.